

# SAFETY DATA SHEET

## ( SDS )

This safety data sheet complies with the requirements of:  
Regulation (EC) No. 1907/2006 and Regulation (EC) No. 1272/2008, (EU) No. 453/2010

Revision Date 23-Mar-2016

WAI2 - EGHS - EUROPEAN

Revision Number 2

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

#### 1.1. Product Identifier

**Product Name** Copper (Soluble)  
**Product No** AC4029-AMP  
**Pure substance/mixture** Mixture

#### 1.2. Relevant identified uses of the substance or mixture and uses advised against

**Recommended Use** Use as laboratory reagent  
**Uses advised against** No Information available

#### 1.3. Details of the supplier of the safety data sheet

**Manufacturer, Importer, Supplier** Thermo Orion Inc. (Part of Thermo Fisher Scientific, Inc.)  
Water Analysis Instruments  
22 Alpha Road  
Chelmsford, MA 01824, USA  
1-978-232-6000

**E-mail address** [wai.techservbev@thermofisher.com](mailto:wai.techservbev@thermofisher.com)

**Made in** USA

**1.4. Emergency telephone number** 24 Hour Emergency Phone Number  
CHEMTREC®  
Within USA and Canada: 1-800-424-9300  
Outside USA and Canada: 1-703-527-3887  
(collect calls accepted)

**SECTION 2. HAZARDS IDENTIFICATION****2.1. Classification of the substance or mixture****Classification - Mixture****Classification according to Regulation (EC) No. 1272/2008 [CLP]**

This mixture is classified as not hazardous according to regulation (EC) 1272/2008 [GHS]

**2.2. Label elements****Signal Word**

None

**Precautionary Statements**

P202 - Do not handle until all safety precautions have been read and understood

**2.3. Other hazards**

No information available

**SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS****3.1. Substances**

| Component        | Chemical Formula         | EC-No.            | CAS-No     | Weight % | CLP Classification - Regulation (EC) No 1272/2008                       | REACH Reg. No            |
|------------------|--------------------------|-------------------|------------|----------|-------------------------------------------------------------------------|--------------------------|
| Water            | No information available | EEC No. 231-791-2 | 7732-18-5  | 70 - 80% |                                                                         | No information available |
| Ammonium Acetate | No information available | EEC No. 211-162-9 | 631-61-8   | 20 - 30% |                                                                         | No information available |
| Sodium Bisulfite | No information available | EEC No. 231-548-0 | 7631-90-5  | 0 - 10%  | Acute Tox. 4 (H302) (EUH031)                                            | No information available |
| Ammonium Oxalate | No information available | EEC No. 214-202-3 | 1113-38-8  | 0 - 10%  | Acute Tox. 4 (H302)<br>Acute Tox. 4 (H312)                              | No information available |
| Acetone          | No information available | EEC No. 200-662-2 | 67-64-1    | 0 - 10%  | Eye Irrit. 2 (H319) (EUH066)<br>STOT SE 3 (H336)<br>Flam. Liq. 2 (H225) | No information available |
| Bathocuproine    | No information available | EEC No. 258-111-7 | 52698-84-7 | 0 - 10%  |                                                                         | No information available |

**Note** \*The exact percentage (concentration) of composition has been withheld as a trade secret

Full text of H- and EUH-phrases: see section 16

**SECTION 4: FIRST AID MEASURES****4.1. Description of first aid measures**

|                                   |                                                                                                                                                                                                                                                                                          |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>             | Use first aid treatment according to the nature of the injury. For further assistance, contact your local Poison Control Center. Show this safety data sheet to the doctor in attendance.                                                                                                |
| <b>Eye Contact</b>                | In case of eye contact, remove contact lens and rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Obtain medical attention.                                                                                                                       |
| <b>Skin Contact</b>               | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. If symptoms persist, call a physician.                                                                                                                                             |
| <b>Inhalation</b>                 | Move to fresh air. If breathing is difficult, give oxygen. Get medical attention if symptoms occur.                                                                                                                                                                                      |
| <b>Ingestion</b>                  | Clean mouth with water and drink afterwards plenty of water. Do not induce vomiting. Call a physician or Poison Control Center immediately.                                                                                                                                              |
| <b>Protection of First-aiders</b> | Use personal protective equipment. See section 8 for more information. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. |

**4.2. Most important symptoms and effects, both acute and delayed**

**Most important symptoms/effects** No information available

**4.3. Indication of any immediate medical attention and special treatment needed**

**Notes to Physician** Treat symptomatically

**SECTION 5: FIREFIGHTING MEASURES****5.1. Extinguishing media****Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Unsuitable Extinguishing Media**

No information available

**5.2. Special hazards arising from the substance or mixture**

Thermal decomposition can lead to release of irritating gases and vapors.

**5.3. Advice for firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6: ACCIDENTAL RELEASE MEASURES****6.1. Personal precautions, protective equipment and emergency procedures**

**Personal Precautions** Use personal protective equipment. Evacuate personnel to safe areas.

**6.2. Environmental precautions**

**Environmental Precautions** Beware of vapors accumulating to form explosive concentrations. Vapors can accumulate in low areas.

### 6.3. Methods and material for containment and cleaning up

**Methods for Containment** Prevent further leakage or spillage if safe to do so.

**Methods for Cleaning Up** Soak up with inert absorbent material. Pick up and transfer to properly labeled containers.

### Reference to Other Sections

Refer to protective measures listed in Sections 7 and 8

See Section 8 for information on appropriate personal protective equipment

See Section 12 for additional Ecological Information

See Section 13 for additional waste treatment information

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

#### Advice on safe handling

To avoid risks to human health and the environment, comply with the instructions for use. Wear personal protective equipment. Avoid breathing dust/fume/gas/mist/vapors/spray. Ensure adequate ventilation, especially in confined areas.

#### General hygiene considerations

Handle in accordance with good industrial hygiene and safety practice.

### 7.2. Conditions for safe storage, including any incompatibilities

#### Storage Conditions

Keep container tightly closed in a dry and well-ventilated place. Store at room temperature in the original container. Keep away from direct sunlight.

### 7.3. Specific end use(s)

#### Specific Use(s)

Use as laboratory reagent

#### Risk Management Methods (RMM)

The information required is contained in this Safety Data Sheet.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

| Component                     | European Union                                        | The United Kingdom                                                                                                    | France                                                                                                                                                                                                                     | Spain                                                                             | Germany                                                                                                                                                                                                                                                            |
|-------------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sodium Bisulfite<br>7631-90-5 | -                                                     | STEL: 15 mg/m <sup>3</sup> 15 min<br>TWA: 5 mg/m <sup>3</sup> 8 hr                                                    | TWA / VME: 5 mg/m <sup>3</sup> (8 heures).                                                                                                                                                                                 | TWA / VLA-ED: 5 mg/m <sup>3</sup> (8 horas)                                       | -                                                                                                                                                                                                                                                                  |
| Acetone<br>67-64-1            | TWA: 500 ppm 8 hr<br>TWA: 1210 mg/m <sup>3</sup> 8 hr | STEL: 1500 ppm 15 min<br>STEL: 3620 mg/m <sup>3</sup> 15 min<br>TWA: 500 ppm 8 hr<br>TWA: 1210 mg/m <sup>3</sup> 8 hr | TWA / VME: 500 ppm (8 heures). restrictive limit<br>TWA / VME: 1210 mg/m <sup>3</sup> (8 heures). restrictive limit<br>STEL / VLCT: 1000 ppm. restrictive limit<br>STEL / VLCT: 2420 mg/m <sup>3</sup> . restrictive limit | TWA / VLA-ED: 500 ppm (8 horas)<br>TWA / VLA-ED: 1210 mg/m <sup>3</sup> (8 horas) | TWA: 500 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 1200 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 500 ppm (8 Stunden). MAK<br>TWA: 1200 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 1000 ppm<br>Höhepunkt: 2400 mg/m <sup>3</sup> |
| Component                     | Italy                                                 | Portugal                                                                                                              | The Netherlands                                                                                                                                                                                                            | Finland                                                                           | Denmark                                                                                                                                                                                                                                                            |
| Sodium Bisulfite<br>7631-90-5 | -                                                     | TWA: 5 mg/m <sup>3</sup> 8 horas                                                                                      | -                                                                                                                                                                                                                          |                                                                                   | TWA: 5 mg/m <sup>3</sup> 8 timer                                                                                                                                                                                                                                   |

|                               |                                                                                                                                                                   |                                                                                                                                                     |                                                                                          |                                                                                                                                                              |                                                                        |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Acetone<br>67-64-1            | TWA: 500 ppm 8 ore.<br>Media Ponderata nel<br>Tempo<br>TWA: 1210 mg/m <sup>3</sup> 8<br>ore. Media Ponderata<br>nel Tempo                                         | STEL: 750 ppm 15<br>minutos<br>TWA: 500 ppm 8<br>horas<br>TWA: 1210 mg/m <sup>3</sup> 8<br>horas                                                    | STEL: 2420 mg/m <sup>3</sup> 15<br>minuten<br>TWA: 1210 mg/m <sup>3</sup> 8<br>uren      | TWA: 500 ppm 8<br>tunteina<br>TWA: 1200 mg/m <sup>3</sup> 8<br>tunteina<br>STEL: 630 ppm 15<br>minuutteina<br>STEL: 1500 mg/m <sup>3</sup> 15<br>minuutteina | TWA: 250 ppm 8 timer<br>TWA: 600 mg/m <sup>3</sup> 8<br>timer          |
| <b>Component</b>              | <b>Austria</b>                                                                                                                                                    | <b>Switzerland</b>                                                                                                                                  | <b>Poland</b>                                                                            | <b>Norway</b>                                                                                                                                                | <b>Ireland</b>                                                         |
| Sodium Bisulfite<br>7631-90-5 | -                                                                                                                                                                 | TWA: 5 mg/m <sup>3</sup> 8<br>Stunden                                                                                                               |                                                                                          | TWA: 5 mg/m <sup>3</sup> 8 timer<br>STEL: 10 mg/m <sup>3</sup> 15<br>minutter.                                                                               | TWA: 5 mg/m <sup>3</sup> 8 hr.<br>STEL: 15 mg/m <sup>3</sup> 15<br>min |
| Acetone<br>67-64-1            | MAK-KZW: 2000 ppm<br>15 Minuten<br>MAK-KZW: 4800<br>mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 500 ppm 8<br>Stunden<br>MAK-TMW: 1200<br>mg/m <sup>3</sup> 8 Stunden | STEL: 1000 ppm 15<br>Minuten<br>STEL: 2400 mg/m <sup>3</sup> 15<br>Minuten<br>TWA: 500 ppm 8<br>Stunden<br>TWA: 1200 mg/m <sup>3</sup> 8<br>Stunden | STEL: 1800 mg/m <sup>3</sup> 15<br>minutach<br>TWA: 600 mg/m <sup>3</sup> 8<br>godzinach | TWA: 125 ppm 8 timer<br>TWA: 295 mg/m <sup>3</sup> 8<br>timer<br>STEL: 125 ppm 15<br>minutter.<br>STEL: 295 mg/m <sup>3</sup> 15<br>minutter.                | TWA: 500 ppm 8 hr.<br>TWA: 1210 mg/m <sup>3</sup> 8<br>hr.             |
| <b>Component</b>              | <b>European Union</b>                                                                                                                                             | <b>United Kingdom</b>                                                                                                                               | <b>France</b>                                                                            | <b>Spain</b>                                                                                                                                                 | <b>Germany</b>                                                         |
| Acetone<br>67-64-1            | -                                                                                                                                                                 | -                                                                                                                                                   | Acetone: 100 mg/L<br>urine end of shift                                                  | -                                                                                                                                                            | -                                                                      |

**Derived No Effect Level (DNEL)** No information available

**Predicted No Effect Concentration (PNEC)** No information available

## 8.2. Exposure controls

**Engineering Measures** Showers  
Eyewash stations  
Ventilation systems

## Personal protective equipment

**Eye/face Protection** Wear chemical splash goggles and face shield. If splashes are likely to occur, wear:.  
Goggles.

**Skin and body protection** Wear protective gloves/clothing.

**Respiratory Protection** No protective equipment is needed under normal use conditions. In case of inadequate ventilation wear respiratory protection.

**Environmental exposure controls** No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

**Physical State** Liquid  
**Appearance** Colorless to pale brown  
**Odor** Sharp  
**Odor Threshold** No information available  
**PH Range** 5.5-8.5

| <u>Property</u>                     | <u>Values</u>            | <u>Remarks • Method</u> |
|-------------------------------------|--------------------------|-------------------------|
| <b>Melting point/freezing point</b> | No information available |                         |
| <b>Boiling Point/Range</b>          | 106 °C / 222.8 °F        |                         |
| <b>Flash Point (High in °C)</b>     | No information available |                         |
| <b>Evaporation Rate</b>             | No information available |                         |
| <b>Flammability (solid, gas)</b>    | No information available |                         |
| <b>Flammability Limit in Air</b>    |                          |                         |

|                              |                          |
|------------------------------|--------------------------|
| Upper flammability limit:    | No information available |
| Lower flammability limit:    | No information available |
| Vapor pressure               | No information available |
| Vapor Density                | No information available |
| Specific Gravity             | No information available |
| Water Solubility             | Soluble in water         |
| Solubility in other solvents | No information available |
| Partition coefficient        | No information available |
| Autoignition Temperature     |                          |
| Decomposition Temperature    | No information available |
| Kinematic viscosity          | No information available |
| Dynamic viscosity            | No information available |
| Explosive Properties         | No information available |
| Oxidizing Properties         | No information available |

## **9.2. Other information**

|                  |                          |
|------------------|--------------------------|
| Softening Point  | No information available |
| Molecular Weight | No information available |
| VOC Content(%)   | No information available |
| Density          | No Information available |
| Bulk Density     | No information available |

## **SECTION 10: STABILITY AND REACTIVITY**

### **10.1. Reactivity**

No information available

### **10.2. Chemical stability**

Stable under normal conditions

### **Explosion Data**

|                                  |      |
|----------------------------------|------|
| Sensitivity to Mechanical Impact | None |
| Sensitivity to Static Discharge  | None |

### **10.3. Possibility of hazardous reactions**

None under normal processing

### **10.4. Conditions to avoid**

Extremes of temperature and direct sunlight

### **10.5. Incompatible materials**

No information available

### **10.6. Hazardous decomposition products**

Thermal decomposition can lead to release of irritating gases and vapors

## **SECTION 11: TOXICOLOGICAL INFORMATION**

### **11.1. Information on toxicological effects**

#### **Acute Toxicity**

#### **Product Information**

Product does not present an acute toxicity hazard based on known or supplied information.

|              |                          |
|--------------|--------------------------|
| Inhalation   | No information available |
| Eye Contact  | No information available |
| Skin Contact | No information available |
| Ingestion    | No information available |

**Unknown Acute Toxicity** 23.1 % of the mixture consists of ingredient(s) of unknown toxicity.

The following values are calculated based on chapter 3.1 of the GHS document

|                               |                 |
|-------------------------------|-----------------|
| ATEmix (oral)                 | 26,194.00 mg/kg |
| ATEmix (dermal)               | 84,590.00 mg/kg |
| ATEmix (inhalation-dust/mist) | 7,705.40 mg/L   |

| Component        | LD50 Oral                 | LD50 Dermal | LC50 Inhalation                            |
|------------------|---------------------------|-------------|--------------------------------------------|
| Water            | LD50 > 90 mL/kg ( Rat )   |             |                                            |
| Sodium Bisulfite | LD50 = 1310 mg/kg ( Rat ) |             |                                            |
| Acetone          | LD50 = 5800 mg/kg ( Rat ) |             | LC50 = 50100 mg/m <sup>3</sup> ( Rat ) 8 h |

**Skin Corrosion/Irritation** No information available

**Serious eye damage/eye irritation** No information available

**Sensitization** No information available

**Mutagenic Effects** No information available

**Carcinogenic effects** No information available

**Reproductive Effects** No information available

**STOT - single exposure** No information available

**STOT - repeated exposure** No information available

**Aspiration hazard** No information available

## SECTION 12. ECOLOGICAL INFORMATION

### 12.1. Toxicity

24.1% of the mixture consists of components(s) of unknown hazards to the aquatic environment

| Component        | Freshwater Algae | Freshwater Fish                                                                                                                                               | Water Flea                                                                                            |
|------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Ammonium Acetate | -                | LC50: = 1.06 mg/L, 48h (Cyprinus carpio)                                                                                                                      | -                                                                                                     |
| Sodium Bisulfite | -                | LC50: = 240 mg/L, 96h static (Gambusia affinis)                                                                                                               | EC50: = 119 mg/L, 48h (Daphnia magna)                                                                 |
| Acetone          | -                | LC50: = 8300 mg/L, 96h (Lepomis macrochirus)<br>LC50: 6210 - 8120 mg/L, 96h static (Pimephales promelas)<br>LC50: 4.74 - 6.33 mL/L, 96h (Oncorhynchus mykiss) | EC50: 12600 - 12700 mg/L, 48h (Daphnia magna)<br>EC50: 10294 - 17704 mg/L, 48h Static (Daphnia magna) |

### 12.2. Persistence and degradability

No information available

### 12.3. Bioaccumulative potential

No information available

| Component | log Pow |
|-----------|---------|
| Acetone   | -0.24   |

**12.4. Mobility in soil**

No information available

**12.5. Results of PBT and vPvB assessment**

No information available

**12.6. Other adverse effects**

No information available

**Endocrine Disruptor Information**

No information available

**SECTION 13. DISPOSAL CONSIDERATIONS****13.1. Waste treatment methods****Waste from Residues / Unused Products**

Disposal should be in accordance with applicable regional, national and local laws and regulations.

**Contaminated Packaging**

Improper disposal or reuse of this container may be dangerous and illegal.

**SECTION 14: TRANSPORT INFORMATION****IMDG/IMO**

|                                                                               |                          |
|-------------------------------------------------------------------------------|--------------------------|
| 14.1 UN-No                                                                    | Not Regulated            |
| 14.2 Proper Shipping Name                                                     | Not Regulated            |
| 14.3 Hazard Class                                                             | Not Regulated            |
| 14.4 Packing Group                                                            | Not Regulated            |
| 14.5 Marine Pollutant                                                         | Not Applicable           |
| 14.6 Special Provisions                                                       | None                     |
| 14.7 Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code | No information available |

**ICAO**

|                           |                |
|---------------------------|----------------|
| 14.1 UN-No                | Not Regulated  |
| 14.2 Proper Shipping Name | Not Regulated  |
| 14.3 Hazard Class         | Not Regulated  |
| 14.4 Packing Group        | Not Regulated  |
| 14.5 Environmental hazard | Not Applicable |
| 14.6 Special Provisions   | None           |

**IATA**

|                           |                |
|---------------------------|----------------|
| 14.1 UN-No                | Not Regulated  |
| 14.2 Proper Shipping Name | Not Regulated  |
| 14.3 Hazard Class         | Not Regulated  |
| 14.4 Packing Group        | Not Regulated  |
| 14.5 Environmental hazard | Not Applicable |
| 14.6 Special Provisions   | None           |

**SECTION 15: REGULATORY INFORMATION****15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture**

European Union



Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work

**International Inventories**

|               |                 |
|---------------|-----------------|
| USINV         | Complies        |
| CANINV        | Complies        |
| EINECS/ELINCS | Complies        |
| ENCS          | Does not Comply |
| IECSC         | Does not Comply |
| KECL          | Does not Comply |
| PICCS         | Does not Comply |
| AICS          | Does not Comply |

USINV/ TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

CANINV/ DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

AICS - Australian Inventory of Chemical Substances

**15.2. Chemical safety assessment**

A Chemical safety assessment according to regulation (EC) No. 1907/2006 is not required

**SECTION 16: OTHER INFORMATION****Key or legend to abbreviations and acronyms used in the safety data sheet****Legend - SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |

**Prepared By** Environmental, Health and Safety

**Prepared For** Thermo Fisher Scientific Inc.

**Issue Date** No information available

**Revision Date** 23-Mar-2016

**Reason for revision** SDS sections updated.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006**

**Disclaimer**

IMPORTANT: The information contained in this SDS is correct to the best of our knowledge as of the issue date (or subsequent revision date, if any), and is to be used only as a guide. This SDS does not constitute a guarantee (express or implied) of any kind and we make no warranties of any kind as to the accuracy or completeness of the information contained herein or the merchantability or fitness of the product or this information for a particular purpose. It is the responsibility of each individual buyer/user to determine the suitability of this information and the product for its intended purposes. Product sales are subject to Thermo Fisher Scientific's standard terms and conditions of sale. This information relates only to the designated product as shipped and may not be valid if the product is used in combination with any other materials or is not used in accordance with our instructions, or is altered in any way. It is the responsibility of the buyer/user to ensure that its activities comply with all applicable government requirements. Since conditions of use of the product are not under direct control of Thermo Fisher Scientific, it is the duty of the buyer/user to determine the necessary conditions for the safe use of the product. Thermo Fisher Scientific will not be liable for any injuries or damages resulting from handling, use, misuse or contact with the product.

**End of Safety Data Sheet**